Hosting on Render: Complete Guide

Step-by-Step Instructions • Part 1: Prerequisites & GitHub Push

100% FREE TIER

🎯 **Goal of this Guide**

Take your local **Baseline Security Demo** application running on your computer and make it available 24/7 on the internet with a public, secure HTTPS link (e.g. `https://baseline-security-demo.onrender.com`) using Render's free tier.

1. What You Need Before You Start

You only need two free accounts. No credit card or paid subscription is required:

SERVICE	PURPOSE	WHERE TO SIGN UP
GitHub	Stores your application code securely in the cloud.	github.com (Free)
Render	Hosts your Node.js application and provides a free HTTPS website.	render.com (Free, sign in with GitHub)

2. How the App Works on Render (Simple Understanding)

This project was built with **zero external runtime dependencies**. It uses Node.js v24 core modules (`node:http`, `node:sqlite`, `node:fs`). This means:

- Render doesn't need to download hundreds of heavy npm packages.
- There are zero build errors, compile steps, or native C++ driver requirements.
- Render deploys it in under **45 seconds**.
- SQLite stores the database automatically inside `data/baseline_demo.sqlite`.

3. Step-by-Step: Uploading Your Code to GitHub

1 Create a New Repository on GitHub

1. Go to github.com and sign in.
2. In the top right corner, click **+** → **New repository**.
3. Name it: `baseline-security-demo` (leave it **Public**).
4. **Important:** Do NOT check "Add a README" (your project already has all files!).
5. Click the green **Create repository** button.

2 Push Your Files from PowerShell

Open **PowerShell** in your project folder (`c:\Users\HP\OneDrive\Desktop\Hackkhers`) and run:

```
# 1. Link your local project to your newly created GitHub repository
git remote add origin https://github.com/<YOUR-GITHUB-USERNAME>/baseline-security-demo.git

# 2. Ensure your branch is named main
git branch -M main

# 3. Push all files to GitHub
git push -u origin main
```

✅ When the push finishes, refresh your GitHub repository page. You will see all project files (`server.js`, `public/`, `render.yaml`, etc.) uploaded and ready!

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Step-by-Step Instructions • Part 2: Deploying on Render Dashboard

RENDER SETUP

4. Step-by-Step: Deploying on Render (The Easiest Method)

3 Connect GitHub to Render

1. Open dashboard.render.com and sign in (choose *Sign in with GitHub*).
2. In the top right, click the blue **New +** button and select **Web Service**.
3. Select **Build and deploy from a Git repository** and click **Next**.
4. Find your `baseline-security-demo` repository in the list and click **Connect**.

4 Fill Out Web Service Settings

Fill out the service creation form exactly as shown in this table:

FIELD NAME	WHAT TO ENTER	WHY
Name	`baseline-security-demo`	Sets your public URL: `baseline-security-demo.onrender.com`
Region	Select closest (e.g. *Oregon* or *Frankfurt*)	Physical server datacenter.
Branch	`main`	The branch where your code lives.
Root Directory	*(Leave blank)*	Files are in root of the project.
Runtime	`Node`	Uses Node.js runtime environment.
Build Command	`echo 'Ready'` *(or leave blank)*	Zero external build steps required.
Start Command	`node server.js`	Starts the server and serves the dashboard.
Instance Type	**Free** ($0 / month)	Render's 100% free hosting tier.

Under Advanced Settings:

Scroll down and click **Advanced**:

- **Health Check Path:** Enter `/api/health` (Render tests this to confirm your app is alive).
- **Environment Variable:** Key: `NODE_ENV` • Value: `production`

Now click the blue "Create Web Service" button!

5 Watch the Deployment Complete

Render will open a live terminal. In less than 1 minute, you will see:

```
==> Starting service with 'node server.js'
================================================================
  BASELINE SECURITY DEMO — Comparator Server
  Fictional baseline workflow for controlled product comparison
----------------------------------------------------------------
  Health API:  http://0.0.0.0:10000/api/health
  Database:    SQLite (data/baseline_demo.sqlite)
================================================================
==> Your service is live 🎉
```

🎉 Render will display your live public link at the top (e.g. `https://baseline-security-demo.onrender.com`)!

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Step-by-Step Instructions • Part 3: Verification, Free Tier & Troubleshooting

TESTING & FAQS

5. How to Test Your Live Website (Acceptance Checklist)

Open your new Render public URL in any browser or phone and test these 5 features:

#	FEATURE	WHAT TO VERIFY
1	**Health API**	Visit `https://<app>.onrender.com/api/health` → Returns `{"status":"ok", "version":"1.0.0"}`.
2	**Overview Page**	Verify the light slate theme, KPI cards, and the 2-agent architecture pipeline diagram.
3	**Live Scan Flow**	Click **Run New Baseline Scan** → Select *Sample 1: Web App API Gateway* → check authorization → click **Start 2-Agent Scan**. Watch the live 2-stage execution timeline!
4	**Download Reports**	Go to **Reports** tab → click **HTML**, **JSON**, and **CSV**. All 3 download immediately.
5	**Comparison Lab**	Go to **Comparison** → click **Load Standardized Lab Suite (N=10)** → verify dynamic Precision and Recall calculations.

6. Understanding Render's Free Tier (Important Notes)

⏳ **1. Inactivity Sleep (Spindown)**

On Render Free tier, if nobody visits your website for **15 minutes**, Render puts the server to sleep to save power. When someone clicks your link again, it takes about **30 to 45 seconds** to wake up (called a "cold start"). Once awake, it runs at full speed.

💾 **2. SQLite Database Behavior**

Render Free tier instances use container storage. If Render restarts your free server, the SQLite database re-initializes back to the clean demo seed data automatically. **This is ideal for presentations** because every demo starts with clean, predictable demo data!

🔄 **3. Automatic Updates on Every Git Push**

Whenever you make changes to your code, commit them, and run `git push`, Render automatically detects the commit and redeploys your live website without needing to touch Render!

7. Quick Troubleshooting Reference

QUESTION / ISSUE	SOLUTION
Git asks for password during push	GitHub deprecated account passwords for command-line pushes. Use a GitHub **Personal Access Token (classic)** with `repo` scope, or push using GitHub Desktop or VS Code.
Render says "Build Failed"	Make sure **Build Command** is empty or `echo 'Ready'`, and **Start Command** is `node server.js`.
How do I share the demo link?	Copy the URL from the top of your Render dashboard (e.g. `https://baseline-security-demo.onrender.com`). Anyone in the world can open it on desktop, tablet, or phone with zero login required.